Movement into and out of cells

Mark scheme and worked solutions for 20 questions: 10 multiple choice, 10 structured.

1 · Part 1

Core | Diffusion | [1]

Answer: B

Explanation

Diffusion is the net movement of particles down a concentration gradient (higher to lower concentration) caused by their random movement. Option A describes active transport. Option C describes osmosis only, which is one special case of diffusion. Option D has the direction the wrong way round: random movement causes net movement from higher to lower concentration.

2 · Part 1

Core | Diffusion | [1]

Answer: D

Explanation

The energy for diffusion comes from the kinetic energy of the random movement of molecules and ions; it is a passive process. Respiration (A) provides energy for active transport, not diffusion. Protein carriers (B) are used in active transport. Light (C) is not the energy source for diffusion.

3 · Part 1

Core | Diffusion | [1]

Answer: A

Worked solution

Compare one factor at a time. Temperature: A and B are warmer. Concentration: A and C have the stronger dye. Surface area: A and D have all six faces exposed. Only A is best in every factor, so the most dye diffuses into A.

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4 · Part 1 Core Osmosis [1]

Answer: C

Explanation

Osmosis is the diffusion of water through a partially permeable membrane. Option A is wrong because it is water, not solute, that moves. Option B is wrong because osmosis is passive and needs no energy from respiration. Option D is wrong because osmosis happens in all cells, including animal cells; the cell wall only stops plant cells bursting.

5 · Part 1 Core Osmosis [1]

Answer: A

Explanation

The solution inside the tubing is more concentrated than the distilled water, so water moves into the tubing by osmosis. The mass increases and the sucrose solution inside becomes more dilute (its concentration decreases). Options C and D wrongly show water leaving. Option B is wrong because adding water dilutes the solution.

6 · Part 1 Core Osmosis [1]

Answer: D

Worked solution

No change in mass means the potato cells neither gained nor lost water overall. Look for where the values change from positive (gained water) to negative (lost water): this happens between 10 % and 15 %.

7 · Part 1 Extended Osmosis [1]

Answer: B

Explanation

The more dilute solution, X, has the higher water potential. Water moves by osmosis from higher water potential to lower water potential, so the net movement is from X to Y. Options A and C have the direction wrong. Option D has the right direction but the wrong reason: X has the higher, not the lower, water potential.

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8 · Part 1 Core Osmosis [1]

Answer: C

Explanation

Plants are supported by the pressure of water inside the cells pressing outwards on the cell walls. When the plant loses more water than it gains, this pressure falls and the soft tissues droop. Option A is wrong because cell walls do not become thinner. Option B is wrong because the plant is short of water, so cells lose water, not gain it; plant cells also do not burst because of the cell wall. Option D is wrong because lack of water does not make membranes fully permeable.

9 · Part 1 Core Active transport [1]

Answer: D

Explanation

Glucose is moving from a lower concentration (gut) to a higher concentration (cells), against the concentration gradient. This is active transport, which uses energy from respiration. Diffusion (A, B) only causes net movement down a concentration gradient. Kinetic energy of the molecules (C) powers diffusion, not active transport.

10 · Part 1 Extended Active transport [1]

Answer: A

Explanation

Sodium ions move into the gill cells against the concentration gradient, so this is active transport. Protein carriers in the membrane move the ions using energy from respiration; when respiration stops, no energy is released and uptake stops. Option B is wrong because diffusion does not need energy from respiration and would move ions out, not in. Option C is wrong because stopping respiration does not change the kinetic energy of the ions. Option D is wrong because osmosis is the movement of water, not ions.

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11 · Part 1 Core Diffusion [6]

Mark scheme

(a) **[2]**

- net movement of particles from a region of higher concentration to a region of lower concentration / down a concentration gradient (1)
- as a result of their random movement (1)

ALLOW molecules or ions for particles

(b)(i) **[2]**

- oxygen moves from the pond water into the organism (1)
- carbon dioxide moves out of the organism into the pond water (1)

(b)(ii) **[1]**

- cell membrane (1)

REJECT cell wall

(c) **[1]**

- oxygen is needed for (aerobic) respiration, so the cells could not release enough energy (1)

Worked solution

(a) Diffusion is the net movement of particles from where they are more concentrated to where they are less concentrated, caused by their random movement.

(b) Oxygen is more concentrated in the pond water (9) than inside the organism (2), so it diffuses in. Carbon dioxide is more concentrated inside the organism (6) than in the water (1), so it diffuses out. Both gases pass through the cell membrane.

(c) The organism uses oxygen for aerobic respiration, which releases the energy its cell needs. Without oxygen it cannot release enough energy and dies.

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12 · Part 1 Core Diffusion [6]

Mark scheme

(a) **[2]**

- as temperature increases, the time taken to become colourless decreases (1)
- correct data quote with units, e.g. 420 s at 15 °C and 200 s at 45 °C (1)

ALLOW time roughly halves between 15 °C and 45 °C

(b) **[2]**

- at higher temperature the (acid) particles have more kinetic energy / move faster (1)
- so they diffuse into the agar faster / the rate of diffusion increases (1)

(c) **[2]**

- any two from (1 each):
- size / volume / surface area of the cubes
- concentration of the acid
- concentration of indicator in the agar
- type / concentration of agar

IGNORE amount of acid (volume is excluded); same temperature

Worked solution

(a) The higher the temperature, the shorter the time. For example, the cube took 420 s at 15 °C but only 200 s at 45 °C.

(b) The energy for diffusion comes from the kinetic energy of the particles. At a higher temperature the acid particles move faster, so they diffuse into the agar more quickly and reach the indicator sooner.

(c) To make it a fair test, only temperature should change. Keep the cube size, the acid concentration and the indicator and agar the same.

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13 · Part 1 Core Diffusion [7]

Mark scheme

(a) **[2]**

- area of one face = 2 × 2 = 4 (cm^2) (1)
- P = 6 × 4 = 24 cm^2 (1)

ALLOW 24 with no working for 2 marks

(b) **[1]**

- Q = 3 : 1 (1)

ECF (answer to (a)) ÷ 8, written as a ratio to 1

(c) **[2]**

- the 1 cm cube has the largest surface area : volume ratio / largest surface area compared with its volume (1)
- the distance from the surface to the centre is shortest / only 0.5 cm (1)

(d) **[2]**

- a large organism has a small surface area : volume ratio / not enough surface for its volume (1)
- the distance to the inner cells is too large, so diffusion would be too slow to supply them (1)

Worked solution

(a) A cube has six square faces. Each face is 2 cm × 2 cm = 4 cm^2, so the total surface area is 6 × 4 = 24 cm^2.

(b) Surface area : volume = 24 : 8, which simplifies to 3 : 1.

(c) As the cube gets bigger, the ratio falls from 6 : 1 to 1.5 : 1. The smallest cube has the most surface for each cm^3 of agar, so the solution can enter over a relatively large area. The centre is also only 0.5 cm from the surface, so the solution has the shortest distance to diffuse.

(d) A large organism is like the large cube: its surface area is small compared with its volume, and many cells are far from the surface. Diffusion over long distances is too slow to supply all these cells with oxygen.

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14 · Part 1 Core Osmosis [4]

Mark scheme

(a) **[1]**

- (soluble) products of digestion dissolve in water so they can be absorbed / enzymes and their substrates are dissolved in water so the reactions can happen (1)

(b) **[1]**

- waste substances such as urea are dissolved in water and removed in urine (1)

ALLOW toxins / excess ions for urea

(c) **[1]**

- dissolved substances are carried in a liquid, e.g. glucose / nutrients / hormones in blood plasma, or mineral ions in xylem / sucrose in phloem (1)

(d) **[1]**

- osmosis (1)

Worked solution

Many substances dissolve in water, so water lets them be moved and used. In digestion, the small soluble molecules made from food are dissolved in water, which allows them to be absorbed. In excretion, urea is dissolved in water and removed from the body in urine. In transport, substances such as glucose and hormones are carried dissolved in blood plasma. Water itself moves into and out of cells through the cell membrane by osmosis.

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15 · Part 1 Core Osmosis [6]

Mark scheme

(a) **[2]**

- change in mass = 14.7 − 12.0 = 2.7 (g) and (2.7 ÷ 12.0) × 100 (1)
- 22.5 % (increase) (1)

ALLOW 23 %; +22.5 %

(b) **[3]**

- any three from (1 each):
- water moves into the bag by osmosis
- from distilled water / dilute solution to the more concentrated sucrose solution / down a water concentration gradient
- through the partially permeable tubing
- sucrose molecules are too large to pass out of the tubing

ALLOW from higher water potential to lower water potential

(c) **[1]**

- the solutions inside and outside are the same concentration, so there is no net movement of water (1)

Worked solution

(a) The mass increased by 14.7 − 12.0 = 2.7 g. Percentage change = (2.7 ÷ 12.0) × 100 = 22.5 %. Check: 22.5 % of 12.0 g is 2.7 g, and 12.0 + 2.7 = 14.7 g.

(b) Distilled water outside the bag is more dilute than the 30 % sucrose solution inside. Water moves by osmosis through the partially permeable tubing, from the dilute side into the concentrated solution. Sucrose cannot pass out, so the bag gains water and its mass increases.

(c) In bag R both liquids are the same, so water molecules move in and out at the same rate. There is no net movement of water and no change in mass.

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16 · Part 1 Extended Osmosis [7]

Mark scheme

(a)(i) **[1]**

- turgid (1)

(a)(ii) **[1]**

- plasmolysed (1)

ALLOW plasmolysis

REJECT flaccid alone

(b) **[3]**

- the sodium chloride solution has a lower water potential than the cell contents / vacuole (1)
- water leaves the cells by osmosis through the partially permeable (cell) membrane (1)
- vacuole and cytoplasm shrink, so the membrane pulls away from the cell wall (1)

ALLOW turgor pressure is lost

(c) **[2]**

- water enters the cells by osmosis because the distilled water has a higher water potential than the cell contents (1)
- the cells become turgid again / turgor pressure increases, and the cell wall stops them bursting (1)

ALLOW the membrane and cytoplasm are pushed back against the cell wall

Worked solution

(a) Cells in distilled water take in water until the vacuole presses the cytoplasm against the wall. These cells are turgid. In strip Y the membrane has pulled away from the wall, which is called plasmolysis; the cells are plasmolysed.

(b) The concentrated salt solution has a lower water potential than the inside of the cells. Water moves by osmosis from the higher water potential (cell) to the lower water potential (solution), through the partially permeable membrane. The vacuole and cytoplasm lose water and shrink, so the membrane pulls away from the cell wall.

(c) Distilled water has a higher water potential than the cell contents, so water moves back into the cells by osmosis. The vacuoles swell and push the cytoplasm against the wall. Turgor pressure builds up and the cells become turgid; the strong cell wall stops them bursting.

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17 · Part 1 Core Osmosis [5]

Mark scheme

(a) **[2]**

- change = 46 − 50 = −4 (mm) and (−4 ÷ 50) × 100 (1)
- −8 % (1)

ALLOW 8 % decrease

ALLOW 8 % if the working shows the decrease (46 − 50)

(b) **[2]**

- water entered the carrot cells by osmosis (1)
- (pressure of) water inside the cells pressed outwards on the cell walls (making the cells firm and supporting the strip) (1)

ALLOW cells became turgid

(c) **[1]**

- any value from 2.4 % to 3.0 % (1)

Worked solution

(a) The length went from 50 mm to 46 mm, a change of −4 mm. Percentage change = (−4 ÷ 50) × 100 = −8 %. Check: 8 % of 50 mm is 4 mm, and 50 − 4 = 46 mm.

(b) In the 0 % solution (water), water moved into the carrot cells by osmosis. The water inside the cells pressed outwards on the cell walls, so the cells were firm and the strip did not bend.

(c) No change means a final length of 50 mm. The strip grew to 51 mm in 2 % and shrank to 48 mm in 4 %, so 50 mm lies between these. 50 mm is one third of the way from 51 mm to 48 mm, so the estimate is about 2 + (2 ÷ 3), which is about 2.7 %.

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18 · Part 1 Extended Osmosis [7]

Mark scheme

(a) **[3]**

- soil water has a higher water potential than the cell sap / cytoplasm of the root hair cell (1)
- water moves into the root hair cell by osmosis (1)
- through the partially permeable (cell) membrane (1)

ALLOW from a dilute solution to a more concentrated solution

(b) **[2]**

- the salt lowers the water potential of the soil water, to below that of the root cells (1)
- so water does not enter / water leaves the root cells by osmosis, so the cells lose turgor (and the plant wilts) (1)

(c) **[2]**

- in both, water enters by osmosis because distilled water has a higher water potential than the cell contents (1)
- red blood cells have no cell wall, but the plant cell wall resists the pressure / stops the plant cell bursting (1)

Worked solution

(a) Soil water is usually a dilute solution, so it has a higher water potential than the cell sap inside the root hair cell. Water moves by osmosis from the higher water potential (soil) to the lower water potential (cell) through the partially permeable cell membrane.

(b) When salt builds up, the soil water becomes more concentrated and its water potential falls. If it falls below the water potential of the root cells, water no longer enters the roots and may even leave them by osmosis. The plant cannot replace water it loses, its cells lose turgor and the plant wilts, even though the soil is wet.

(c) Distilled water has the highest water potential, so water moves into both kinds of cell by osmosis. A red blood cell has only a cell membrane, which stretches until it bursts. A plant cell has a strong cell wall; as the cell swells, the wall pushes back (turgor pressure) and stops it bursting.

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19 · Part 1 Core Active transport [5]

Mark scheme

(a) **[1]**

- potassium (1)

(b) **[1]**

- the concentration of potassium is higher inside the cells (400) than in seawater (10), so it must move against the concentration gradient (1)

(c) **[1]**

- respiration (1)

ALLOW aerobic respiration

(d) **[2]**

- at lower temperature respiration is slower / the enzymes for respiration are less active (1)
- so less energy is released (for active transport), so the ions are moved in more slowly (1)

Worked solution

(a) and (b) Potassium is 10 units in seawater but 400 units inside the cells. To keep moving into the cells it must go from a lower to a higher concentration, against the gradient, which only active transport can do. Sodium is more concentrated in seawater, so it would not need active transport to enter.

(c) Active transport uses energy released by respiration.

(d) Respiration is controlled by enzymes, which work more slowly at 5 °C than at 20 °C. Less energy is released, so less is available for active transport and potassium is taken up more slowly.

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20 · Part 1 Extended Active transport [6]

Mark scheme

(a) **[1]**

- 25 (times) (1)

(b) **[1]**

- the concentration is higher inside the cells than in the soil water, so diffusion would move nitrate ions out of the cells, not in (1)

ALLOW they must move against the concentration gradient

(c) **[1]**

- protein carriers (in the membrane) (1)

(d) **[2]**

- waterlogged soil contains less oxygen, so the root cells carry out less (aerobic) respiration (1)
- so less energy is released for active transport (1)

(e) **[1]**

- the plant could not take up enough nitrate (because the soil concentration is lower than in the cells), so it could not make enough amino acids / proteins (1)

ALLOW either idea alone for the mark: cannot absorb nitrate ions from this soil, or cannot make amino acids / proteins for growth

Worked solution

(a) 10 ÷ 0.4 = 25, so the concentration inside the cells is 25 times greater. Check: 0.4 × 25 = 10.

(b) Diffusion only causes net movement from a higher to a lower concentration. Here the higher concentration is inside the cell, so diffusion would carry nitrate out, not in.

(c) Protein carriers in the cell membrane move the ions across, using energy.

(d) When soil is waterlogged, water fills the air spaces, so the roots get less oxygen. Less aerobic respiration takes place in the root hair cells, so less energy is released for active transport, and uptake drops from 30 to 6 units.

(e) Active transport is the only way the plant can take in nitrate from this soil, because there is less nitrate outside the cells than inside. Without it, the plant would lack nitrate and could not make enough amino acids and proteins for growth.

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